

Breaking The Bottleneck

I just want a faster PC!

Nimish Chandiramani

It's a shame how even an ultra-fast PC with a dual-core processor and oodles of RAM still takes ages to copy a file from a CD to your hard disk. Or how Windows XP refuses to load faster than 25 seconds. And how your favourite game will run at 50 frames per second at the highest setting, but will still have you twiddling your thumbs while you wait for the next level to load. You're probably getting the drift here—it's all very well to make leaps and bounds in technology, but what's the point if you still have a few rubberbands holding you back? It's like running to a car in the rain without getting wet, only to find out that the guy with the keys is still 50 metres away.

The Fat Slug

The first thing that needs to go is the hard disk. While data can zoom along on your motherboard's system bus at speeds approaching 10 Gbps, the SATA-II hard disk interface ambles on at a leisurely 300 MBps at peak per-

formance (around 150, realistically). Clearly something needs to be done. While we pray for the death of the hard disk and its impending replacement by new technologies such as carbon nanotubes (read about alternatives to the hard disk in *Digit* March 2006), work is already being done to do away with the hard disk even in today's PCs.

Well, to be honest, the idea has been there for ages—use solid-state storage. RAM chips are capable of data transfer rates of 6 to 8 Gbps, and it makes more sense to have data in there rather than sitting on a hard disk platter. There was just one catch—RAM is volatile, so any data on it is erased as soon as you stop supplying power to it. The answer is to supply a small battery that will keep the data alive even when the computer is switched off, and *voila!* No more waiting for programs to load into memory—they'll already be there!

At Computex 2005, Gigabyte introduced iRAM, which is a PCI card with DIMM slots for RAM chips, which can be used as permanent storage. It fools the PC into believing that it's actually a hard disk using the SATA II interface. And therein lies the problem. You see, while you're still gaining the advantage of greater reliability thanks to no moving parts, you don't gain speed, because the SATA II interface will limit you to a maximum of 300 MBps!

A much better (read much, much more drool-worthy) implementation is what the guys at Atom Chip (www.atomchip.com) have come up with—they base their super laptops completely on their quantum-optical RAM. The latest SG220-2 laptop sports 2 TB (yes, that's Terabytes!) of quantum-optical memory. Try to relate that to your sit-

uation today—a 2,000 GB hard disk? Wow! 2,000 GB of *system memory*? {Faint}

Take Me Into The Light

There is only one ultimate speed—the speed of light. We're not going to rest till all technology moves at said ultimate speed, which is why we're so obsessed with making fibre optics work everywhere possible. Thanks to fibre optic cables, it's actually possible to have an Ethernet network that works at 10 Gbps. But what good can we get out of this if the copper connections on the Ethernet cards and motherboards only support 4 or 5 Gbps?

Enter Intel, which has been researching the possibility of bringing fibre optics to the motherboard. Instead of the copper tracks we have today, components will be connected to each other using little fibre optic lines. This will oust copper—which can only do 10 Gbps at its peak—for fibre optics will take these transfer speeds to anywhere between 30 and 60 Gbps. And one hallowed day, fibre optics will be used *inside* chips as well—imagine over a billion transistors connected optically without even needing a heatsink!

The catch (for there must be one) is that while fibre optic materials are getting cheaper, the equipment to convert electrical signals into optical signals and vice versa is dauntingly expensive. As of today, there is no way that any of the above dreams can be commercially viable.

Intel to the rescue again—in 2004, they tested a new microchip, called a modulator, which works at 1 GHz, is made of silicon, and is a far sight cheaper than other modulators, which were based on much more expensive materials like gallium arsenide.

It's still going to take around 10 years or so, but the super-fast silicon-optical chip isn't a hilarious fantasy any more.

And now the ultimate question: What does one do with a PC like that? ■

nimish_chandiramani@thinkdigit.com

Illustration Pravin Warhokar